Trainings

General Information

All K38 and K50 engines are equipped with an engine barring device. The purpose of this device is to provide a method of barring the engine to adjust the valves and injectors. The barring device is provided because the capscrews on the front of the crankshaft that are **not** capable of the force required to rotate the engine are **not** accessible in many of the applications. K38 and K50 engines have the marks used to position the engine for adjusting the valves and injectors located on the flywheel and flywheel housing, near the barring device mounting location, and on the vibration dampers. Some engines have two barring devices.

The barring device contains a gear that engages the flywheel ring gear. The device also contains a spring that will push the barring device gear from the flywheel ring gear when the engine is operating. An external cotter clip is used to prevent the barring device gear and shaft from moving and contacting the ring gear when the engine is in operation.

Engines that contain a **wet** type flywheel housing **must** have a gasket installed between the barring device and the flywheel housing to prevent oil leaks. Engines with a **dry** type flywheel housing do **not** require this gasket.

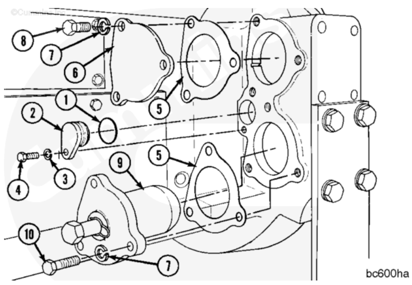
Remove

Note : Only engines equipped with wet type flywheel housing have the gaskets (5).

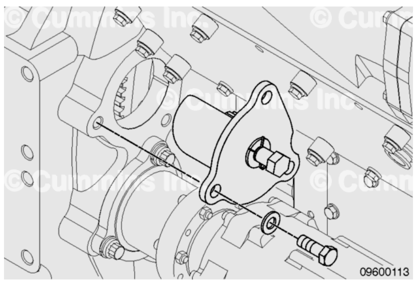
Remove the barring device (9), the starter bore cover (6), and the access hole cover (2).

Remove the gaskets and seals.

Discard the gaskets and seals.



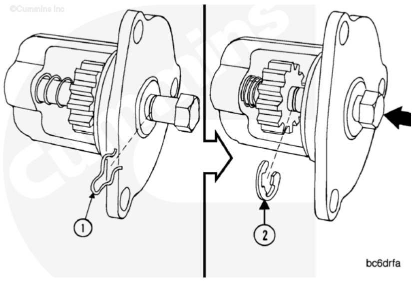
Remove the three capscrews and the engine barring device from the flywheel housing



Disassemble

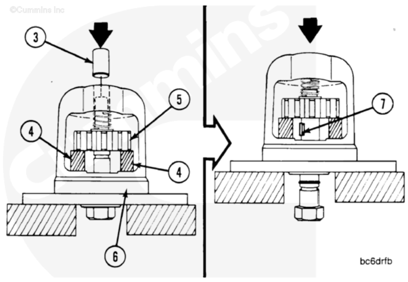
Remove the external cotter clip (1).

Push the shaft in. Remove the e-ring (2).



⚠ CAUTION ⚠

Do not attempt to push the shaft out without spacers or supports between the gear and the housing. The key in the shaft will damage the housing.



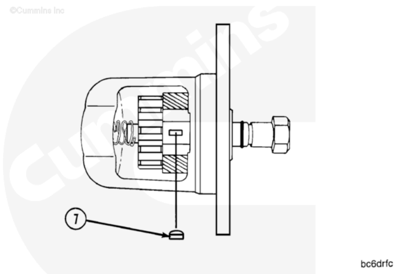
Use an arbor press and a mandrel (3). The mandrel **must** pass through the shaft bore in the housing.

Put supports (4) between the gear (5) and the housing (6). The supports **must** be a **minimum** of 21 mm [0.813 in] long to allow the key to clear the gear and **not** contact the housing.

Note : Steel 3.18 mm [0.125 in] pipe couplings have been used successfully for the supports (4).

Push the shaft until the key (7) is exposed.

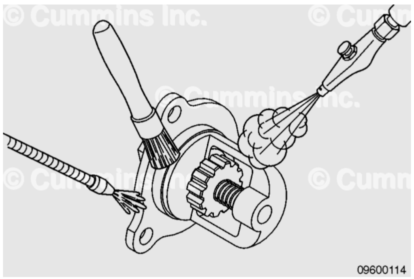
Remove the key (7). Remove the supports.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

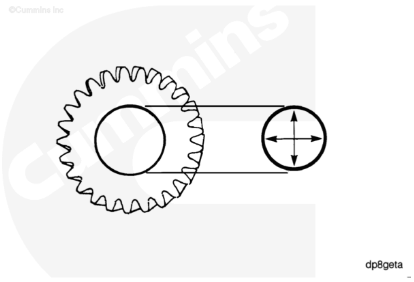
⚠ CAUTION ⚠

Do not use a solvent that will damage the o-ring on the shaft. An oil leak can result.

Use solvent to clean the parts. Dry with compressed air.

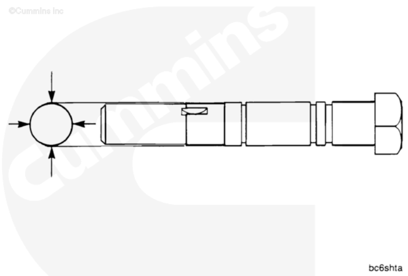
Check the gear teeth for damage or wear.

Measure the inside diameter of the gear.



Barring Device Gear Inside Diameter		
mm		in
18.77	MIN	0.739
18.80	MAX	0.740

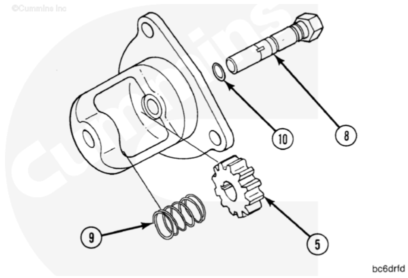
Measure the gear step of the shaft.



Barring Device Shaft Outside Diameter		
mm		in
18.821	MIN	0.7410
18.834	MAX	0.7415

Assemble

Install the o-ring (10) on the shaft. The o-ring **must** be in the **second** groove from the hex end of the shaft.

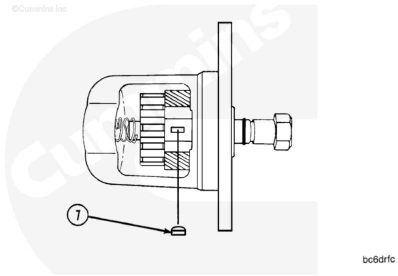


Install the gear (5) and spring (9) in the housing. The spring will secure the gear in the housing.

Use clean engine oil. Lubricate the o-ring and shaft.

Install the shaft (8) through the housing and in the gear until the step touches the gear.

Note : The supports used to remove the gear from the shaft can be used between the gear and the housing to allow the installation of the key.



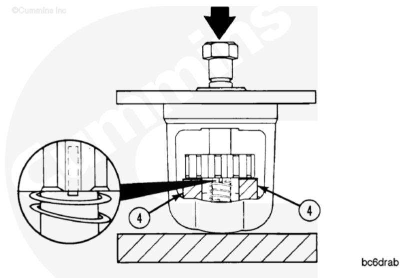
Push the shaft **in** until the keyway is exposed.

Install the key (7) in the shaft.

Align the keyway in the gear with the key.

⚠ CAUTION ⚠

Make sure the key is aligned with the keyway in the shaft before pushing the shaft in the gear or damage to the parts will result.

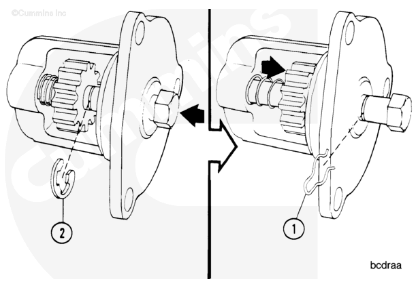


Use an arbor press. Use the supports (4) as illustrated.

Push the shaft into the gear until the end of the key is even with the side of the gear.

Remove the spacers.

Push the shaft **in** until the groove for the e-ring is exposed.

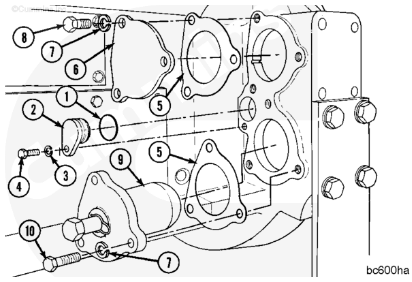


Install the e-ring (2).

Push the shaft **out** until the e-ring touches the housing. Install the external cotter clip (1).

Install

Note : The illustrations show the parts being installed on the left bank of the flywheel housing. Some engines have the parts installed on the right bank of the engine. The parts shown are always installed on the opposite side of the flywheel housing from the starting motor.



Install the o-ring seal (1) on the access hole cover (2).

Use vegetable oil. Lubricate the o-ring seal.

Install the access hole cover in the bore in the flywheel housing.

Install the lock washer (3) and the capscrew (4).

Torque Value: 9 n•m [80 in-lb]

Note : The gaskets (5) must be installed on a wet type flywheel housing. The gaskets are not required on a dry type flywheel housing.

Install the parts.

1. Gasket
2. Cover Plate
3. Lock washer (three)
4. Capscrews (three)

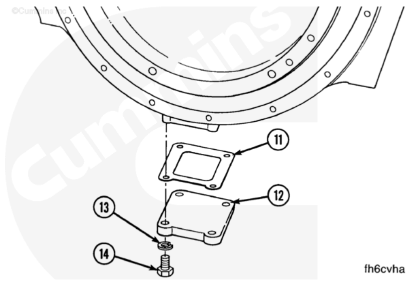
Torque Value: 60 n•m [45 ft-lb]

Install the parts.

1. Gasket
2. Engine Barring Device
3. Lock washer (three)
4. Capscrews (three)

Torque Value: 185 n•m [135 ft-lb]

Note : The cover plate has pipe threads for a wet type flywheel housing drain.



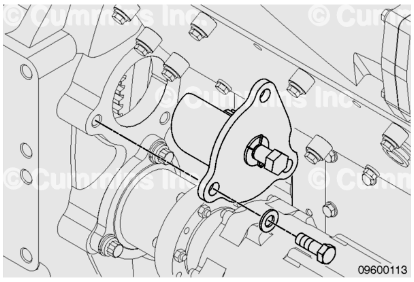
Install the parts

1. Gasket
2. Plate
3. Lock washers (four)
4. Capscrews (four)

Torque Value: 40 n•m [30 ft-lb]

Install the engine barring device, three lock washers and capscrews.

Torque Value: 180 n•m [135 ft-lb]



Last Modified: 28-Mar-2019